Nobel Prize has been awarded for work in the foundations of quantum mechanics.

After the onset of World War II, Heisenberg led a German government program to develop nuclear weapons. What Heisenberg actually thought about the Nazis, and whether he truly tried as hard as possible to push the weapons program forward, are matters of some historical dispute. It seems that, like a number of other Germans, Heisenberg was not fond of the Nazi Party, but preferred a German victory in the conflict to the prospect of being run over by the Soviets. There is no evidence that he actively worked to sabotage the nuclear bomb program, but it is clear that his team made very little progress. In part that must be attributed to the fact that so many brilliant Jewish physicists had fled Germany as the Nazis took power.

As impressive as matrix mechanics was, it suffered from a severe marketing flaw: the mathematical formalism was highly abstract and difficult to understand. Einstein's reaction to the theory was typical: "A veritable sorcerer's calculation. This is sufficiently ingenious and protected by its great complexity, to be immune to any proof of its falsity." (This from the guy who had proposed describing spacetime in terms of non-Euclidean geometry.) Wave mechanics, developed immediately thereafter by Erwin Schrödinger, was a version of quantum theory that used concepts with which physicists were already very familiar, which greatly helped accelerate acceptance of the new paradigm.

Physicists had studied waves for a long time, and with Maxwell's formulation of electromagnetism as a theory of fields, they had become adept at thinking about them. The earliest intimations of quantum mechanics, from Planck and Einstein, had been away from waves and toward particles. But Bohr's atom suggested that even particles weren't what they seemed to be.

In 1924, the young French physicist Louis de Broglie was thinking about Einstein's light quanta. At this point the relationship between photons and classical electromagnetic waves was still murky. An obvious thing to contemplate was that light consisted of both a particle and a wave: particle-like photons could be carried along by the well-known electromagnetic